MIL-S-43963A(GL)
7 January 1983
SUPERSEDING
MIL-S-43963(GL)
25 March 1977

MILITARY SPECIFICATION

SHIRTS, WOMAN'S: JUNIOR R.O.T.C.

This specification is approved for use by Natick Research and Development Laboratories, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers polyester/cotton, durable press white shirts.

1.2 Classification. The shirts shall be of two types in the following sizes (see 6.2).

Type I - Long sleeves

Type II - Short sleeves

Schedule of sizes

6	14
8	16
10	18
12	20

2. APPLICABLE DOCUMENTS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8410

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* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-B-871	-	Button, Sewing Hole, and Button, Staple (Plastic)
UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-T-43548	-	Thread, Polyester, Cotton-Covered and Rayon-Covered
MIL-C-43836	-	Cloth, Interlining, Non-Woven
MIL-C-43479	-	Cloth, Broadcloth, Polyester and Cotton

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment
MIL-STD-1492	-	Provisions for Evaluating Quality of Men's Shirts

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

TEXTILE COLOR CARD ASSOCIATION

Standard Color Card of America

(Application for copies should be addressed to the Textile Color Card Association of the United States, Inc., 24 East 38th Street, New York, NY 10016.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

(Application for copies of the AATCC Manual should be addressed to the American Association of Textile Chemists and Colorists, Research Triangle Park, P.O. Box 12215, Durham, NC 27709.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide samples. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

3.2 Materials (see 6.5).

* 3.2.1 Basic material. The basic material for the shirts shall be white polyester/cotton conforming to type I, class 4, of MIL-C-43479.

3.2.2 Interlining. The material for interlining the collar and cuffs shall be a continuous filament, non-woven polyester cloth conforming to type I, class 1 of MIL-C-43836.

3.2.2.1 Bias binding. When option to use bias binding in collar setting operation is selected, binding may be cut from basic material to a length and width as specified in table II, operation 7 d and e.

3.2.3 Thread. The thread shall be polyester, cotton-covered and rayon-covered, size 70, 2 ply, conforming to MIL-T-43548. The color of the thread shall be white AH, C.A. 66050.

3.2.4 Braid. The braid for the top button loop shall be white, 100 percent filament nylon tubular construction made on a 4 line machine using size Z-33 nylon thread (3 ends of 100 denier each). The braid shall conform to the following physical requirements:

Diameter, inches	1/16 to 3/32
Carriers (min)	16
Ends per carrier (min)	1
Picks per inch (min)	38
Breaking Strength, lbs. (min)	55
Yards per lb. (min)	600

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3.2.5 Labels. Each shirt shall have an identification, a size and an instruction label conforming to type VI of DDD-L-20. The fastness to laundering requirements of DDD-L-20 shall apply.

3.2.5.1 Identification label. The identification label shall conform to class 1.

3.2.5.2 Size label. The size label shall conform to class 2.

3.2.5.3 Instruction label. The instruction label shall conform to class 3 and shall contain the following information:

HAND OR MACHINE WASH
Use warm water and detergent.
DO NOT USE chlorine bleach or starch.
Drip dry or tumble dry at medium heat.
Remove immediately.
Press at synthetic (medium heat).

3.2.5.4 Combination label. A combination identification and instruction label may be used. The combination label shall conform to the requirements of the class 3 label.

3.2.6 Buttons. The buttons for front closure and cuffs shall be type II, style 25, class K, 20 or 21-line, polyester, White BA, Cable No. 62031, conforming to V-B-871.

3.3 Design. The shirts have a plain front with cut-on facings, five button front closure, yoke back, convertible collar and a straight hemmed bottom. The type I shirt has long sleeves pleated into a two button cuff. The type II shirt has short sleeves with hems. (See figures 1 and 2).

3.4 Patterns. Standard patterns will be furnished by the Government and, unless otherwise indicated, provide an allowance of 1/4 inch for all seams. The standard patterns shall not be altered in any way and are to be used only as a guide for cutting the contractor's working patterns, which shall be identical to the standard patterns.

3.4.1 Pattern parts. The components of the shirt shall be cut from the material as specified below, in accordance with the pattern parts indicated.

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TABLE I. List of pattern parts

Material	Pattern nomenclature	Cut parts
Polyester/cotton cloth	Front	2
	Back	1
	Yoke	1
	Sleeve (type I)	2
	Sleeve binding (gusset)	2
	Cuff (type I)	2
	Sleeve (type II)	2
	Collar	2
Spun-bonded polyester cloth	Collar interlining (use collar pattern)	1
	Cuff interlining (type I)	2

* 3.5 Construction. The construction shall conform in all respects to the requirements specified in table II herein. Figures 1 and 2 are furnished solely for guidance and information. Should variation from the written document appear in figures 1, and 2, the written document shall govern.

* 3.5.1 Stitches, seams and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table II. Seam allowances shall be maintained with seams sewn so that no-raw edges, run-offs, pleats, puckers or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the shirt.

* 3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewed.

* 3.5.1.1.1 Repairs of type 301 stitching.

a. When thread breaks, skipped stitches, run-offs, or bobbin runouts occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

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b. Except for prestitching, thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

* 3.5.1.2 Type 401, 515, 516, and 519 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.5.1.1.1a. and b. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn. 301 stitching may be used to repair 401 stitching.

* 3.5.2 Bartacking. Bartacking shall be 1/2 ± 1/16 inch long, 1/8 ± 1/32 inch wide and shall contain 28 stitches (minimum). Bartacks shall be free from thread breaks and loose stitching.

3.5.3 Buttonholes. The buttonholes shall be straight-cut and the finished cut length shall be 1/2 to 9/16 inch long. The stitching shall be securely caught in the fabric and the ends shall be securely tacked. When a purl-type buttonhole is used, the purling shall be on the outside of the shirt. All loose thread ends shall be trimmed to within 1/8 inch of the tack. Additive treatments or materials used to secure the ends of the stitching or to stiffen the fabric prior to stitching the buttonholes are prohibited.

3.5.4 Automatic stitching. Automatic machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of the stitching.

3.5.5 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the shirts.

3.6 Manufacturing operations requirements. The shirt shall be made in accordance with operation requirements specified in table II. The contractor is not required to follow exact sequence of operations listed unless otherwise specified. Any additional basting or holding stitches used to facilitate manufacture are permissible provided the thread does not show on the finished shirt.

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NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cut shirt. a. Cut shirt in strict accordance with patterns furnished which show size, directional lines and notches for assembly. Directional lines shall be placed in the warp direction. b. Cut all basic material parts from same piece of material. The fronts (edge of facing) may be cut on the selvage, otherwise add 1/4 to 3/8 inch seam allowance for hemming (see operation 7.a). The front patterns laid on the uneven selvage edge shall be blocked and reset before cutting. Cut neck bias binding (when applicable) in accordance with 7.d. and 7.e. c. Cut the button loop braid of sufficient length to comply with operation 7.b.						
*2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1492 shall be removed from production and replaced with non-defective and properly matched components.						
*3.	Component marking. a. Mark, and ticket all components of the shirt except those parts cut from ends to ensure correct size, proper assembly, and uniform shade throughout the shirt.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
*3.	Component marking. (cont'd)						
	b. The use of ink pad numbering machine, rubber stamp or pencil will be acceptable provided the numbers do not show through the outside of the shirt and the numbers are covered by the seam allowance.						
	c. Any method of marking may be used except: 1. Metal fastening devices. 2. Sew-on type tickets. 3. Adhesive type tickets which leave traces of adhesive on the material upon removal of the ticket.						
4.	Make cuffs.						
	Finished appearance. The cuffs shall finish smooth and flat with square corners and the buttonholes correctly positioned.						
	a. Fold cuffs face to face, with the interlining next to the top cuff. Seam ends, catching the interlining in the stitching.	301 or 401	SSa-1 SSa-1	10-14 10-14	70 70	70 70	
	b. Trim corners, if necessary, turn and work out points and ends.						
	c. Make two horizontal buttonholes on each cuff to correspond with marks on pattern. The end of the buttonholes shall be 1/2 inch from front edge.	304		52-56 per buttonhole including tack	70	70	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make collar. Finished appearance. The collar shall finish smooth and flat, uniform in shape and with points of equal length.						
	a. Stitch top collar to interlining along the neck edge. The stitching shall not be exposed on the finished collar.	301	SSa-1	6-8	70	70	
	b. Stitch top collar (interlining attached) to undercollar, matching notches and distributing fullness of top collar. Trim points, turn collar and work out points and edges. On the finished collar, seam allowances at ends shall finish against the under collar.	301 or 401	SSa-1 SSa-1	10-14 10-14	70 70	70 70	
6.	Attach yoke. Finished appearance. The finished yoke shall be without twists or puckers. There shall be no fullness between the plies of material.						
	a. Stitch yoke fronts and backs with a safety stitch folding pleats on fronts and back as indicated by marks on pattern. The folded edge of the pleats shall face the outside of the shirt.	515 or 516 or 519	SSa-2 SSa-2 SSa-2	10-14 10-14 10-14	70 (chain stitch) 70 (over edge)	70 70	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Join collar and attach size label.						
	a. If facing is not cut on selvage, turn under raw edge 1/4 to 3/8 inch and stitch 1/16 to 1/8 inch from turned edge. Shoulder edge of facing may also be turned under and stitched at this time.	301	EFa-1	10-14	70	70	
	b. Fold front facing to outside on fold line indicated by notches on pattern. A button loop to fit the collar button shall be inserted in the right step at neck and securely caught in the stitching. Insert collar between front and facing with collar end at collar notch, and with shoulder edge of facing turned down (if not hemmed in operation 7.a), stitch collar to neck from front folded edge to shoulder seam. Shoulder seam allowance shall be turned towards the back. Clip neck seam allowance at shoulder seamline; continue stitching under collar to back of neck and reverse procedure at other end of collar.	301	LSbj-1	10-14	70	70	
	c. Clip gorge seam allowances. Turn facing to right side and work out steps to corners. Stitch shoulder edge of facing to back of shirt, through shoulder seam allowances or if shoulder edge of facing was hemmed in operation 7.a, tack ends with 1/2 inch straight line tack to back of shirt, through shoulder seam allowances. Turn under top collar raw edge and stitch top collar to neck, closing collar, catching size label centered (1/2 inch tolerance).	301 and 301 or 301 (linetack)	LSb-1	10-12 10-12 10-12	70 70 70	70 70 70	
	- or -						
	When alternate construction to use bias binding is selected:						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Join collar and attach labels. (cont'd)						
	d. Fold front facing to outside on fold line indicated by notches on pattern. A button loop to fit the collar button shall be inserted in the right step at neck and securely caught in the stitching. Insert collar between front and facing with collar end at collar notch, and with shoulder edge of facing edge of facing turned down (if not hemmed in operation 7.a) and with bias binding positioned on top collar so that ends of binding overlaps facing ends 1/2 inch or more. With raw edges of binding, collar, and neck even, stitch from front folded edge of facing to opposite folded edge of facing.	301	SSa-1	10-14	70	70	
	e. Clip gorge seam allowance. Turn facing to finished position and work out steps to corners. Turn binding under, bind seam allowances and stitch through facing end across neck to opposite facing end, with stitching overlapping facings 1/2 ± 1/8 inch. Stitching shall be 1/16 to 1/8 inch from turned edge of binding catching size label (centered 1/2 inch tolerance). The binding shall finish 5/16 ± 1/16 inch in width.	301		10-14	70	70	
*8.	Attach labels. Sew instruction and identification labels 1/8 inch ± 1/16 inch from edges on all four sides to the inside as follows:						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
*8.	Attach labels. (cont'd)						
	a. Position the instruction label on left facing with bottom edge of label 1/2 to 3/4 inch from finished bottom, and stitch through facing only on all four sides. Stitching shall not be through printing.	301	LSbj-1	10-14	70	70	
	b. Position identification label on left facing with the bottom edge of label no more than 1/4 inch above instruction label. Stitch through facing only on all four sides. Stitching shall not be through printing. Bottom of identification label may be caught in the top row of stitching of instruction label.	301	LSbj-1	10-14	70	70	
	c. Combination label (when used) shall be positioned as specified for the instruction label (see operation 8.a) and stitch through facing only on all four sides. Stitching shall not be through printing.	301	LSbj-1	10-14	70	70	
9.	Set sleeves.						
	Finished appearance. The sleeves shall be set in the armholes with fullness equally distributed and without pleats or puckers.						
	a. Place a tension (shirr) stitch at cap of sleeve, between notches, to distribute fullness evenly. (If sleeve setting machine has an automatic device for distributing cap fullness, this operation may be omitted.	301		10-14	70	70	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Set sleeves. (cont'd)						
	b. With shoulder seam allowance turned toward back, stitch sleeve to armhole with a safety-stitched seam.	515 or	SSa-2	10-14	70 (chain stitch)	70	
		516 or 519			70 (over edge)	70	
	c. The type II shirt sleeves may be pre-hemmed prior to setting sleeves with the hem folded to the outside and raw edge turned under. Stitch hem 1/16 to 3/32 inch from turned edge. The hem shall finish 1 ± 1/8 inch wide.	301	EFb-1	10-14	70	70	
10.	Join sides.						
	a. Join side seams and sleeve seams in one operation with a safety stitched seam. The armhole seam, at underarm, shall not be staggered more than 1/4 inch.	515 or	SSa-2	10-14	70 (chain stitch)	70	
		516 or 519			70 (over edge)	70	
	b. If sleeves are pre-hemmed (operation 9.c) turn underarm seam allowance toward back of sleeve and place a horizontal bartack or line tack, 1/8 to 3/16 inch from bottom of sleeve, on the underarm seam, catching the seam allowance and end of the safety stitching:	1/2 inch Bartack or		28 per bartack	70	70	
		301 Linetack		28 per tack	70	70	

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					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Make sleeve openings - type I shirt.						
	Turn raw edges of the binding (gusset) to the inside, insert the edge of sleeve opening and seam. The binding shall finish 1/2 to 5/8 inch and be smooth and flat. The sleeve opening shall measure 4 ± 1/4 inches long exclusive of cuff.	301	BSc-1	10-14	70	70	
12.	Join cuffs to sleeves - type I shirt.						
	Finished appearance. The ends of cuffs shall not finish beyond the edges of the sleeve openings.						
	a. Seam the underside of cuff to bottom of sleeve with raw edges even, catching the pleats in the seam and with the bound edge of the sleeve opening turned to the inside at buttonhole end of cuff and caught in the stitching.	301 or 401	LScf-2(a) LScf-2(a)	10-14 10-14	70 70	70 70	
	b. Turn cuff to finished position, and with raw edge on top side of cuff turned under, stitch to sleeve 1/16 to 1/8 inch from turned edge.	301	LScf-2(b)	10-14	70	70	
13.	Hem sleeve - type II shirt (if not pre-hemmed).						
	Turn up hem of sleeve with raw edge folded under 1/4 to 3/8 inch and underarm seam turned toward back. Stitch hem through sleeve 1/16 to 3/32 inches from turned edge. Finished hem shall measure 1 ± 1/8 inches wide.	301	EFb-1	10-14	70	70	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
14.	Hem shirt.						
	Turn bottom of fronts and back of shirt with raw edges turned in 1/4 inch and stitch through shirt, with the hem extending across facings. Hem shall finish 1/4 to 5/16 inch wide.	301	EFb-1	10-14	70	70	
15.	Make buttonholes.						
	Finished appearance. Buttonholes shall be straight and clean from frayed material threads. The stitching shall be securely caught in the fabric and the ends shall be securely tacked.						
	Make four horizontal buttonholes on right front as indicated by marks on pattern with the cut of the buttonhole 3/4 ± 1/8 inch from front edge. Buttonholes shall be vertically aligned and equally spaced. The center of the cut of the bottom buttonhole shall be 5 ± 1/4 inches from bottom edge of hem on finished shirt.	304		52-56 per buttonhole including tack	70	70	
16.	Sew buttons.						
	a. Sew a 20 or 21-line button at the top of the left front to correspond with the loop at top of right front. Collar ends shall be centered equidistant from center of shirt front and shall not overlap.	301 or 101 or Hand		12-16 per button	70	70	
	b. Sew four 20 or 21-line buttons on the left front to correspond with buttonholes on right front. Center of buttons shall be 3/4 inch from front edge.	301 or 101 or Hand		12-16 per button	70	70	

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					NEEDLE	BOBBIN/ LOOPER	COVER
16.	Sew buttons. (cont'd)						
	c. Sew two 20 or 21-line buttons on the outside of each cuff at the back end, to correspond with the buttonholes. The center of the buttons shall be 1/2 inch from the edge.	301 or 101 or Hand		12-16 per button	70	70	
17.	Clean shirt.						
	Trim thread ends to 1/4 inch maximum, remove loose threads, spots, stains and size markings.						
18.	Press shirt.						
	a. Press shirt smooth and flat.						
	b. Button shirt and fold for packaging.						

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3.7 Finished measurements. The finished measurements shall conform to measurements for each respective size, as shown in table III.

TABLE III. Finished measurements (inches)

Sizes	6	8	10	12	14	16	18	20	Tolerance inches
Half bust 1/	18	18-3/4	19-1/2	20-1/4	21	21-3/4	22-1/2	23-1/4	± 1/2
Back length 2/	23-3/4	23-7/8	24	24-1/8	24-1/4	24-3/8	24-1/2	24-5/8	± 3/8
Sleeve length 3/ -type I	21-1/2	21-11/16	21-7/8	22-1/16	22-1/4	22-7/16	22-5/8	22-7/8	± 3/8
-type II	6-7/8	7	7-1/8	7-1/4	7-3/8	7-1/2	7-5/8	7-3/4	3/8
1/2 sleeve width 4/ -type II	5-3/8	5-1/2	5-3/4	6	6-1/4	6-1/2	6-3/4	7	± 3/8
Collar 5/	12-1/4	12-3/4	13-1/4	13-3/4	14-1/4	14-3/4	15-1/4	15-3/4	± 3/8

1/ Across front of shirt, at base of armhole, from fold to fold.

2/ Along center back, from neck seam to bottom hem edge.

3/ Along vertical center of sleeve from shoulder seam at top to bottom edge of cuff for type I and bottom hem edge for type II.

4/ Along bottom edge of sleeve hem, from fold to fold.

5/ Along neck seam of collar from one end to opposite end.

3.8 Workmanship. The finished shirt shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for

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the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. In addition to testing provisions contained in subsidiary specifications, testing shall be performed on the component listed in table IV for characteristics noted. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table IV shall be followed. All requirements are applicable to the sample unit. Unless otherwise specified in subsidiary specifications, sampling shall be in accordance with the following:

Lot size	Sample size
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. All test reports shall contain the individual values utilized in expressing the final result. The unit for expressing lot size shall be "yards" and the sample unit for testing the braid shall be 1-1/2 yards.

TABLE IV. Component tests

Component	Characteristic	Requirement paragraph	Test method
Nylon filament braid	Diameter	3.2.4	1/
	Carriers	3.2.4	Visual
	Ends per carrier	3.2.4	Visual
	Picks per inch	3.2.4	Visual
	Breaking strength	3.2.4	4102
	Yards per lb.	3.2.4	4108

1/ The diameter shall be determined at three points at least 9 inches apart by steel rule graduated in thirty-seconds of an inch or by a caliper-gauge and the results averaged.

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* 4.2.2 End item visual examination. The end item shall be examined for the defects listed in table V. The lot size shall be expressed in units of shirts. The sample unit shall be one shirt. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 15.0 for total (major and minor combined) defects.

TABLE V. End item visual defects

Defect	Classification Major	Minor
Pressing		
a. Omitted		X
b. Burned or scorched	X	
c. Sleeves, lapels, or pleats creased		X
d. Front fold-lines not creased		X
Material defects and damages		
a. Shade bar, smash, multiple float, or loose slub	X	
b. Needle chew, cut, tear, or hole	X	
Shaded parts		
Any outside part shaded	X	
Cleanness		
a. Spot or stain clearly noticeable on outside	X	
b. Thread ends not trimmed, loose threads, or shade ticket not removed		X
c. Basting or holding stitching visible on outside of shirt		X
d. Any marking visible on outside of shirt		X
Component and assembly		
a. Any component not as specified	X	
b. Any component part or required operation omitted (unless otherwise classified herein)	X	
c. Any operation not performed as specified (unless otherwise classified herein)		X

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor
Cutting		
Any part not cut in conformance with warp directional lines indicated on patterns or not cut in accordance with document requirements.		
Seams and stitching		
a. Accuracy of seaming:		
(1) Any part of shirt caught in an unrelated operation or stitching	X	
(2) Seam twisted or pleated (unless otherwise classified herein)		X
(3) Ends of stitching not secured as specified		X
b. Gauge of stitching and seam allowances:		
(1) Irregular, i.e. unevenly gaged or corresponding stitchings not uniformly gaged (to be scored only when condition exists along major portion of seam)		X
(2) Width not as specified or not within range of gage specified, or varies more than 1/16 inch when no range is specified		X
c. Stitches, broken or missing:		
(1) All stitching except buttonholes and overedging		
- up to 1/2 inch (inclusive)		X
- more than 1/2 inch	X	
(2) Overedging		
- more than 1 inch		X
d. Skipped stitches (except on buttonholes) or runoffs:		
- 1/4 to 1/2 inch (inclusive)		X
- more than 1/2 inch	X	
e. Raw edge more than 1/4 inch long on outside		X
f. Seam and stitch type:		
(1) Seam or stitch type not as specified		X
(2) Looper thread of 401 stitch type on outside of shirt	X	

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor
Seams and stitching (cont'd)		
g. Stitches per inch (to be scored only when the condition exists for a distance of more than half the length of the seam):		
(1) More than maximum specified		X
(2) Less than minimum specified	X	
h. Stitch tension		
(1) Loose tension resulting in loose seam	X	
(2) Loose tension (exposed loops of needle or bobbin thread)		X
(3) Tight tension (on 301 stitching, stitches break when stain is applied in direction of seam)	X	
(4) Loose tension on bartacks		X
i. Seam not worked out to edge (when required) having a depth between folds of 1/16 inch or more		X
Labels		
a. Size, identification, instruction, or combination identification and instruction (if applicable) missing, incorrect, or illegible	X	
b. Any label (except size) not stitched on all four sides		X
c. Stitching through the printing of any label		X
d. Any label not positioned as specified		X
Buttonholes		
a. Omitted, added, or not specified type	X	
b. Ragged edges or stitching not caught in fabric		X
c. Broken stitch or two or more skipped stitches	X	
d. Tacking omitted on ends		X
e. Purling not on outside		X
Buttons		
a. Missing, broken, defective, or unsecurely sewn	X	
b. Stitching not locked at end of cycle (tug at loose end of thread, when accessible to determine if it will ravel)	X	

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor
Collar		
a. Points not worked out		X
b. Ends poorly shaped or not uniform in length or shape		X
c. Edges irregular		X
d. Twisted or tight, causing edges to curl up		X
e. Uneven in width by more than 1/8 inch		X
f. Undercollar exposed at collar end by more than 1/16 inch		X
Sleeve		
a. Sleeve fullness not evenly distributed at sleeve cap		X
b. Sleeve tight at armhole causing puckers or pleats on front, back, or sleeve	X	
c. Armhole seam staggered at underarm by more than 1/4 inch		X
d. Sleeve hem twisted, puckered, or pleated (type II - shirt)		X
e. Width of hem less than 7/8 or more than 1-1/8 inches (type II - shirt)		X
Cuff(s) - type I - shirt		
a. Reversed, i.e., buttonhole on underside and button on top side	X	
b. End of cuff finished beyond outer edge of sleeve opening by more than 1/8 inch		X
c. Width less than specified		X
d. Irregular in width by 1/4 inch or more		X
Front		
a. Buttonholes unequally spaced by more than 1/4 inch		X
b. Buttons or buttonholes out of vertical alignment by 1/8 inch or more		X
c. Lapel corner not worked out, having fold of 1/16 inch or more		X
d. Button loop missing from lapel corner or unsecurely sewn		X
e. Front uneven in length at bottom when buttoned, by more than 1/4 inch		X

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor
Front (cont'd)		
f. Fullness on front between buttons and buttonholes		X
g. Cut of two or more buttonholes or center of two or more buttons less than 5/8 inch or more than 7/8 inch from front opening edge	X	
h. Center of cut of bottom buttonhole to bottom of hem less than 4-3/4 inches or more than 5-1/4 inches		X
Yoke		
a. Folded edge of pleat on outside of shirt faced toward center of shirt	X	
b. Seam allowances of fronts and backs not turned toward yoke		X
Bottom hem		
a. Twisted, puckered, or pleated		X
b. Width not as specified		X
c. Hem end extending more than 1/8 inch beyond front edge of shirt		X
Facing		
a. Twisted, puckered, or pulled at collar or shoulder		X
b. Inside of facing not cut on selvage or not hemmed		X

* 4.2.3 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in table III. Any dimension deviating from the specified requirements and sleeves uneven in length by 3/4 inch or more shall be classified as defects. The lot size shall be expressed in units of shirts. The sample unit shall be one shirt. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.4 Packaging inspection. An examination shall be made to determine whether the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flap, improper taping, or inadequate stapling. Open and continuous heat-sealed seams and closure or polyethylene bag. Omission of vent hole in polyethylene bag. Alternative to heat-sealed closure polyethylene bag not provided with tuck or reverse flap. Bulged or distorted container.
Contents	Number of unit packs per shipping container more or less than required.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each shirt shall be neatly folded in accordance with standard commercial practice. Each shirt shall be inserted in a close-fitting clear polyethylene film bag having a thickness of 0.00125 inch (± 25 percent tolerance). The bag shall be formed with heat-sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat-sealed with the heat-seal made as close as possible to the open end. The bag may be fabricated from polyethylene film tubing or sheeting. A 1/4 inch diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape. Alternatively, the final closure of the polyethylene bag may be accomplished by means of a tuck or reverse flap.

* 5.1.2 Commercial. Shirts shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level B or Commercial as specified (see 6.2).

5.2.1 Level B. Sixty shirts of one type and size only, preserved as specified in 5.1, shall be packed in a close-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, variety DW, minimum grade 200 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, minimum grade 200 of

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PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4. Toward the end of the contract or purchase order or when there are less than the required amount per container of the same size, mixed sizes of one type only may be packed within the same container.

* 5.2.2 Commercial packing. Shirts preserved-packaged as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

* 5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

5.3.1 Labels, mixed sizes. Each shipping container packed with mixed sizes of one class only, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED NSN's" plainly printed or stamped thereon and under these words shall be legibly printed or stamped the correct quantity and NSN's contained therein.

6. NOTES

6.1 Intended use. The shirts covered by this document are intended for wear by women in the Junior ROTC Program.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) Type and size required (see 1.2).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Samples. For access to sample of the end item and shade samples, address the contracting activity issuing the invitation for bids.

6.4 Seam appearance. Experience has shown that if the following precautions are observed in using proper components and setting up the sewing machines, seam puckering will be held to an acceptable minimum level.

a. The needles used for all seams should be as fine as possible consistent with the size or diameter of the thread used.

b. The throat plate should be no larger than necessary to permit free entry of the needle.

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c. The finest feed (approximately 18 teeth per inch) possible should be used with a presser foot to exactly match the shape and size of the feed. The face of the presser foot should be highly polished to eliminate roughness and reduce drag.

d. The lightest presser foot presser should be used to prevent stretching of the top ply of material.

e. In the safety stitch machines (stitch types 516, 517 and 519) the upper and lower trimming knives should be kept at peak sharpness to prevent "dragging" of the edge of the fabric.

f. On stitch type 301 sewing, the bobbin tension should be adjusted so that when a full bobbin is placed in the bobbin case and threaded through the tension spring, the bobbin and case shall ride slowly down the thread when the free end of the thread is held in the vertical position. Needle tension should be adjusted after the bobbin tension to provide slightly more needle thread than bobbin thread in the stitch.

g. Thread take-up mechanism should be set on all machines to relieve as much tension as possible when forming the stitch. All thread contacts should be well polished and free from burrs and cuts.

* 6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

* 6.6 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to last previous issue.

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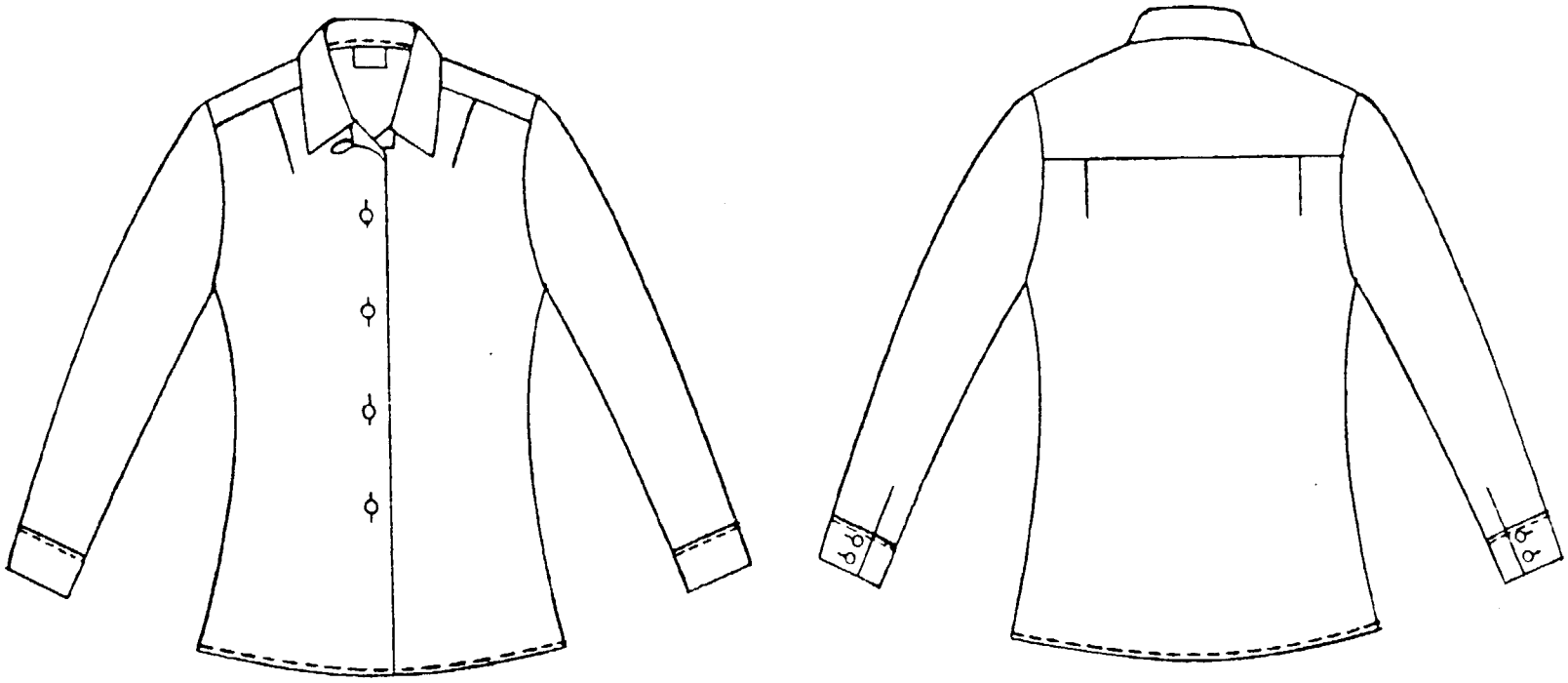
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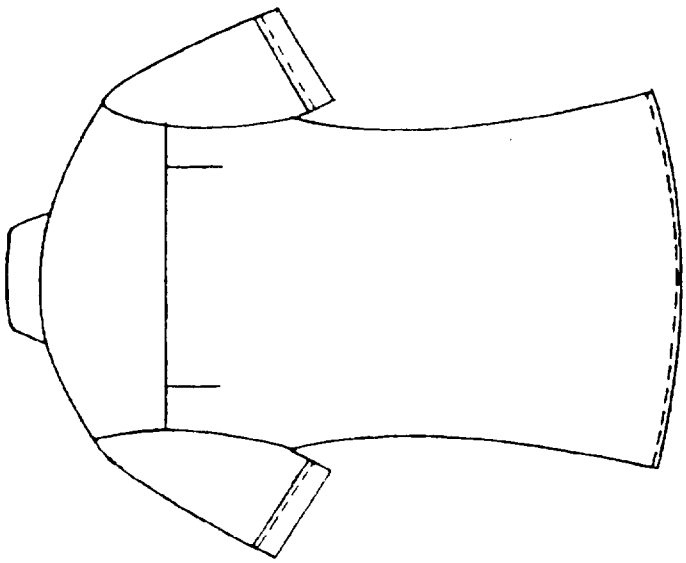


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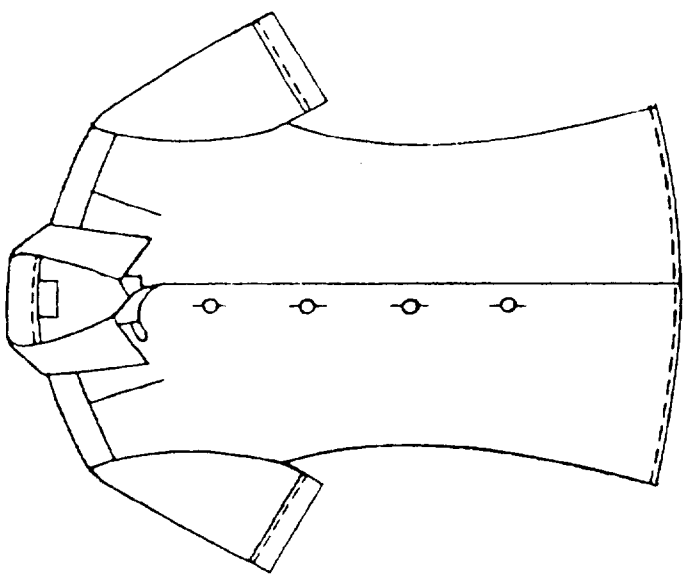
FIG. 1 SHIRTS, WOMAN'S: JUNIOR ROTC TYPE-I LONG SLEEVE

2-1-1378

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BACK



FRONT

FIG. 2 SHIRTS, WOMAN'S: JUNIOR ROTC
TYPE-II SHORT SLEEVE

2-1-379

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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